

RAMP / MEC Scoring – Investigation Task

Wergonic · 17 August 2026

What this is

Several RAMP and MEC numbers disagree with each other across the phone app, the web app and the Word report. We do not currently know which side is right, and most of these have only been argued from memory, never checked against the official RAMP document.

This is an investigation task, not a fix task. Do not change behaviour unless a point below explicitly says to fix it. For every other point, the deliverable is a short written answer: what the code actually does today, what the official source says, and what you would change. We decide after that, you do not need to make any product call yourself.

Start here

Find the official RAMP documentation the scoring was based on. Almost every point below is an open argument that nobody has settled with the source in hand. Write down the page number for each answer so it stops being an opinion.

To check and report back

1. Head bending backwards is measured three different ways

The threshold is -10° in the risk verdict and the Word report, and -5° on the phone and in RAMP. The same worker gets a different answer depending on where you look. Farhad ruled -5 in writing on 3 August; the code was never changed.

Trunk backwards stays at -10° . That one is deliberate – do not merge the two.

There is also a test in the repo claiming -10 was “confirmed by the team”. Find out who wrote it and who “the team” was.

***Answer:** what the official table says, and every place in the code the threshold is written.*

2. Is a RAMP score of 1.5 valid?

Head-backwards is the only one of the six posture scores built with 5 steps instead of 7, and the only one that ever outputs 1.5 or 6. Two other questions already output 0.5, so fractional scores do exist somewhere in our code.

A code comment says someone rewrote this ladder “to match the official RAMP table”. Find out who, and against what.

***Answer:** the official step values for this question, checked one by one, with the page number.*

3. The same score gets two different colours

Two separate pieces of code decide the colour of a score and they disagree. A 1.5 is yellow on the compare-assessments table and green on the single results page. Separately, a score of 1 is green everywhere except question 1.8, where it is yellow.

***Answer:** what the correct colour rule is according to the source. Then fix it so there is one rule used everywhere.*

4. Hand distance ignores the selected task

A RAMP for one task should only look at that task's time. Five of the six posture measures do. Hand distance uses the whole recording instead, so a result labelled as one task is mixed with the rest of the day.

Be careful, this is bigger than it looks: that number is calculated once when the recording is uploaded and stored as a session total. It probably cannot be filtered afterwards, it has to be recalculated.

***Answer:** why it works this way, and an estimate of the change. Do not start before we see it.*

5. Hand distance is a guess presented as a measurement

There is no sensor on the hand – the position is estimated from the arm angles and the length of the arm. The assessor can type the worker's arm and forearm length when running an assessment. MEC uses what they typed. RAMP ignores it and always uses 32 cm and 27 cm, hardcoded, so the two assessments can disagree about the same worker.

***Answer:** confirm this, then fix: make RAMP use the entered lengths, and label the number as an estimate.*

6. Trunk and head percentiles never reach the Word report

They are calculated and stored correctly, then dropped at the last step because the report table only knows how to print arm rows.

***Answer:** confirm, then fix: add forward and backward bending rows for trunk and head. Side bending stays out, Farhad says it is not accurate enough to publish.*

7. A perfect result looks the same as no result

If a worker never bends past the threshold, RAMP questions 1.1 and 1.2 show nothing at all – no score, no colour. That looks exactly like a question where the sensor was missing. Questions 1.3 to 1.6 show a green 0 in the same situation, which is correct.

***Answer:** confirm, then fix: make 1.1 and 1.2 show the green 0 too, and check it is counted in the green total at the top of the page.*

8. RAMP question 1.5 shows the wrong question

The heading is about upper arm posture, hand in or above shoulder height. The question printed underneath asks about bending the upper body forwards or to the side – that is question 1.3's text, copied word for word. So the ergonomist is asked about the back while the score underneath is about the arm. The help text is correct, only the question is wrong.

***Answer:** fix it, in all five languages.*

9. RAMP sub-rows are all numbered 1.1

Inside every RAMP item the rows restart at 1.1, so "1.1" appears under item 1.1, again under 1.2, again under 1.3, and so on. They are not sub-questions, they are what we measured: forward bending, side bending, backward bending.

***Answer:** fix it: name them instead of numbering them.*

10. What is MEC question 11 supposed to measure?

The screen calls it “Neck bending”. The code calculates it as the head angle minus the body angle. The backend's internal name says it should be the head's own angle. Both are defensible – do not pick one.

Same block of code has a second problem: the head and trunk readings are matched by their position in the list instead of by time. If either sensor drops a reading, everything after it compares the head at one moment against the trunk at another, and the extra readings are silently thrown away. Everywhere else in the system matches by time.

***Answer:** document what it does today and what the source expects, and find out why the earlier attempt at fixing the time matching was reverted.*

11. MEC questions 12.1 and 12.2 are dead

They are listed in the backend but never calculated, and nothing on the web side reads them. No screen has a place for them, so no customer has ever seen a blank.

***Answer:** confirm that, and tell us whether to delete them or build them.*

12. The AI section of the report can disappear silently

If the AI call fails, the report is still generated, just without that section.

***Answer:** confirm this is what the code does now, and whether the user gets any indication at all. No change yet.*

Send back one short answer per point. Anything you cannot settle from the source, say so and stop there – do not guess.